

The Topological Origin of Reality: A Unified Framework Deriving Spacetime, Physics, and Computation from Information Constraints

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https://github.com/sferoze/topology_of_the_universe

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Abstract

We propose a unified, background-independent framework deriving the fundamental architecture of reality from a single axiom: the logical instability of absolute nothingness. We argue that the transition from non-existence to existence is necessitated by the logical incoherence of a void devoid of axiomatic constraints. This instability forces a self-relation—a topological folding—which generates distinctions interpreted as information. We demonstrate that persistence of this information requires topological irreversibility (locking). By modeling the balance between connectivity gain and informational cost, we derive a stability threshold $n^* \approx 2.678$, establishing $D = 3$ as the minimal integer dimensionality for stable topological structures (e.g., Trefoil knots) under unconstrained ambient isotopy. This determines the dimensionality of emergent spacetime and defines the fundamental information unit $I = D \cdot 2^D = 24$. This structure constitutes the Information Layer, a compact topological space projecting into 3D spacetime. We establish a unification of physical constants ($c \equiv \hbar = 24$) within this framework's Topological Natural Units (TNU) and provide a rigorous mapping to SI units via dimensional regularization. This derivation yields fundamental scales significantly different from standard Planck units. The framework offers a hypothesis regarding computational complexity (P vs. NP) by leveraging topological adjacency in physical systems, and defines a cyclical cosmology driven by information saturation ($24^{89} \approx 10^{123}$ bits).

1 Introduction: The Crisis of Background Dependence

Contemporary physics faces profound challenges rooted in the assumption of *background dependence*—the reliance on a pre-defined spacetime container within which physical objects exist. This reliance introduces the *substrate problem*: the origin of the fundamental entities themselves.

Theories like Loop Quantum Gravity (LQG) [5, 4], Causal Set Theory, and Digital Physics [23] attempt to address background dependence by discretizing spacetime or assuming fundamental computational rules. We propose a deductive resolution, the "Self-Folding Origin" model, which shifts the ontological focus from objects to the relational topology that defines them, aiming for a truly background-independent account where reality bootstraps itself from logical necessity.

We begin with a foundational axiom addressing Leibniz's question ("Why is there something rather than nothing?"):

Axiom 1 (Instability of Absolute Nothingness). *Absolute nothingness (0), defined as the absence of all distinctions, relations, and axiomatic frameworks, is logically incoherent and therefore unstable.*

This differs from the empty set $\emptyset$ in ZFC set theory, which exists by axiom. "Absolute nothingness" implies the absence even of the axioms that define $\emptyset$. In such a state, the assertion of non-existence is self-referential and paradoxical (akin to the Liar Paradox). The absence of a logical framework prevents the stabilization of the state of nothingness.

From this instability, the Void must resolve itself. It does so through the only operation available in the absence of substance: self-relation, or *folding*. This topological operation is the necessary consequence of the instability, generating the first distinction—the fundamental constraint we identify as information. While this introduces topological operations as primitives, we argue this is a more minimal ontological commitment than presupposing spacetime or substance.

2 The Mechanism of Emergence: The Self-Folding Void

The Void's incoherence forces a default to 'something'. The transition is driven by logical necessity. The Void resolves its instability by relating to itself through folding. This is not a physical process; it is an operation on potentiality itself. The fold spontaneously emerges as a distinction, creating a boundary that separates the undifferentiated Void:

$$\emptyset \rightarrow \emptyset|\emptyset \quad (1)$$

This operation generates a fundamental duality: the boundary (the constraint, the information) and the context (the regions defined relative to the fold). Information requires no physical substrate; the topological operation itself *is* the information [2]. The universe is the totality of these constraints, arising from the recursive self-relation of the Void: $F(\emptyset) = \emptyset|F(\emptyset)$.

3 Topological Persistence and the Fundamental Unit ($I = 24$)

The framework is synthesized in the capstone formula:

$$I = D \times R \quad (2)$$

where I is Total Information, D is the number of Distinctions (depth of folding, or complexity), and R is the number of Relations (configurations) at that depth.

3.1 The Requirement of Topological Locking ($D = 3$)

The value of D is derived from the requirement of persistence. Simple folds ($D = 1, 2$) are topologically trivial (the Unknot) and will spontaneously unfold. Persistence demands a configuration that is topologically locked—an irreversible fold. This corresponds mathematically to the minimal crossing number required for a non-trivial knot, the Trefoil configuration (3_1) [1, 19].

3.2 Mathematical Derivation of Stability

We model stability by balancing the topological advantage gained by folding against the informational cost of maintaining the structure. Let n be the folding depth (corresponding to D). We propose the following scaling relations based on fundamental arguments:

1. **Connectivity Gain** $\propto (n-1)$: Represents the increase in relational complexity compared to the unfolded state ($n = 1$). A linear increase is the most parsimonious assumption for the gain associated with adding discrete distinctions.

2. **Information Cost** $\propto \ln(2n)$: Represents the complexity of encoding the structure. The information required to specify a configuration among $2n$ distinct regions or intersections created by n folds scales logarithmically, following standard information theory principles.

Stability requires the gain to exceed the cost:

$$n - 1 > \ln(2n). \quad (3)$$

To find the critical threshold n^* , we solve the equality $n - 1 = \ln(2n)$ using the Lambert W function, defined as the inverse function of $f(W) = We^W$. We rearrange the equation:

$$e^{n-1} = 2n \quad (4)$$

$$-\frac{1}{2e} = (-n)e^{(-n)} \quad (5)$$

The relevant physical solution corresponding to the stability threshold ($n > 1$) lies on the lower branch (W_{-1}).

$$n^* = -W_{-1}\left(-\frac{1}{2e}\right) \quad (6)$$

Numerically, $n^* \approx 2.678347$ (Verification in Appendix A). Since the folding depth n must be an integer (representing discrete topological operations), the minimal stable configuration requires $n = 3$. Thus, $D = 3$.

3.3 The Fundamental Information Unit

With $D = 3$ established, the relations R represent the exhaustive binary configurations (e.g., fold orientations, chirality) at each distinction: $R = 2^D = 8$. This yields the fundamental information unit:

$$I = 3 \times 8 = 24 \quad (7)$$

4 The Emergence of Spacetime

The requirement for the minimal stable lock ($D = 3$) dictates the dimensionality of the embedding space required for this structure to persist. Spacetime is the emergent geometry necessary to embed these topological constraints.

The critical insight from knot theory is that non-trivial knots (embeddings of 1D loops) are uniquely stabilized in exactly three spatial dimensions under standard ambient isotopy [6].

- **In 3 Dimensions:** Only trivial knots exist.
- **In 3 Dimensions:** Non-trivial topological locks (like the Trefoil) are stable.
- **In ≥ 4 Spatial Dimensions:** Any embedding of a 1-dimensional fold in $\mathbb{R}^{N \geq 4}$ can be trivially unknotted via unconstrained ambient isotopy.

While higher-dimensional knotting exists (e.g., 2-spheres knotted in 4D), the principle of parsimony dictates that the universe emerges in the minimal dimensionality required to stabilize the simplest persistent information structure ($D = 3$) without presupposing external constraints (i.e., unconstrained isotopy). Therefore, the embedding space is exactly three-dimensional. Time emerges as the sequence of these irreversible folding events.

5 The Information Layer and Projection

The $I = 24$ structure exists within the *Information Layer*, a compact topological manifold embodying the totality of folded constraints. Our observable 3D universe is a projection of this higher-complexity layer.

Constraints that appear vastly separated in the 3D projection may be topologically adjacent in the Information Layer. This adjacency underpins the resolution of physical non-locality (entanglement) and provides mechanisms for bypassing computational complexity barriers.

6 Unification of Fundamental Physics and Dimensional Regularization

We derive the fundamental constants of physics within the natural units defined by the $I = 24$ structure. We denote these units as Topological Natural Units (TNU).

6.1 Causality, Action, and Gravity in TNU

The Speed of Light (Causality): The maximum rate of information propagation, c , is defined by the ratio of the minimal spatial distinction (δl) to the minimal temporal distinction (δt).

- $\delta l = D = 3$
- $\delta t = 1/R = 1/8$

$$c_{TNU} = \frac{\delta l}{\delta t} = D \cdot R = I = 24. \quad (8)$$

Planck's Constant (Action): We define the minimal action S_{min} over one complete topological cycle (2π radians), associated with the fundamental information unit I :

$$\hbar_{TNU} = \frac{S_{min}}{2\pi} = \frac{I \cdot 2\pi}{2\pi} = I = 24. \quad (9)$$

The Gravitational Constant (G): We posit a relationship where G scales inversely with the rigidity of spacetime, defined by the cube of the speed of light in these units:

$$G_{TNU} = \frac{1}{c_{TNU}^3} = \frac{1}{24^3} = \frac{1}{13824}. \quad (10)$$

6.2 The Unified View and Dimensional Regularization

This framework reveals the fundamental unity $I \equiv c \equiv \hbar = 24$ in TNU. Information, causality, and the quantum of action are manifestations of the same underlying topological structure.

To connect this model to empirical reality, we must map TNU to standard SI units via dimensional regularization. We define scaling factors for length (κ_L), time (κ_T), and mass (κ_M).

We relate the SI values of the constants to their TNU values:

$$c_{SI} = c_{TNU} \cdot \frac{\kappa_L}{\kappa_T} = 24 \frac{\kappa_L}{\kappa_T} \approx 2.998 \times 10^8 \text{ m/s} \quad (11)$$

$$\hbar_{SI} = \hbar_{TNU} \cdot \kappa_M \frac{\kappa_L^2}{\kappa_T} = 24 \kappa_M \frac{\kappa_L^2}{\kappa_T} \approx 1.055 \times 10^{-34} \text{ J} \cdot \text{s} \quad (12)$$

$$G_{SI} = G_{TNU} \cdot \frac{\kappa_L^3}{\kappa_M \kappa_T^2} = \frac{1}{13824} \frac{\kappa_L^3}{\kappa_M \kappa_T^2} \approx 6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2 \quad (13)$$

This system of equations is well-constrained and yields a unique solution. Solving the system (see Appendix A for derivation and verification) yields the fundamental scales of the topological model:

- $\kappa_L \approx 4.561 \times 10^{-32}$ m (Topological Length Scale)
- $\kappa_T \approx 3.651 \times 10^{-39}$ s (Topological Time Scale)
- $\kappa_M \approx 7.713 \times 10^{-12}$ kg (Topological Mass Scale)

These derived scales differ significantly from the standard Planck scales (e.g., Planck length $\approx 1.616 \times 10^{-35}$ m). This suggests that the fundamental discretization occurs at a much larger length scale than typically assumed, offering a specific and testable prediction of the model.

6.3 Pathway to Formalization: Emergent Physics

General Relativity: Gravity is hypothesized to emerge from the geometry induced by variations in the density of topological constraints. Mass/energy is equivalent to the complexity of topological folding. A pathway toward deriving the Einstein Field Equations involves analyzing how variations in topological complexity density ρ_I affect the emergent geometry, potentially using formalisms akin to Regge calculus or emergent gravity models where the Ricci scalar $R \propto \rho_I$.

Quantum Mechanics: Quantum phenomena reflect the ambiguity inherent in the projection from the Information Layer. Superposition corresponds to the unresolved topology of folds. The evolution of the system can be conceptualized via a path integral formulation where the paths are sequences of topological operations (folding events).

$$Z = \int \mathcal{D}[\text{Topologies}] e^{iS_I/\hbar} \quad (14)$$

where S_I is the action defined by the complexity of the configuration. Measurement adds constraints, resolving the ambiguity (collapse), aligning with Relational Quantum Mechanics [22].

7 Computational Implications: P vs. NP as a Projection Artifact (Hypothesis)

The apparent intractability of NP-complete problems arises from the vast search spaces observed in the 3D projection. This projection obscures constraints that are topologically adjacent in the Information Layer.

We hypothesize that physical systems, when evolving towards stable configurations (e.g., protein folding), utilize the native topology of the Information Layer. By accessing this topology, invalid configurations are eliminated multiplicatively rather than additively.

This collapses the search space exponentially. For example, a 90-bit search (2^{90} possibilities) might be subject to adjacent constraints totaling 2^{80} . The remaining search space is $2^{90}/2^{80} = 2^{10} = 1024$ tractable candidates (Appendix A).

Clarification: This does not constitute a formal proof that $P = NP$ in the abstract sense of computational complexity theory (Turing machines). Rather, it suggests that *physically realized* computation inherently leverages topological shortcuts. The P vs. NP distinction may be an artifact of analyzing problems within the constraints of the 3D projection rather than the native topology.

8 The Imperative of Complexity: Life and Consciousness

The imperative for maximizing information growth (derived from the continuous folding driven by the Void's instability) demands the emergence of increasingly sophisticated self-referential structures.

Growth $\rightarrow$ Exponential Growth $\rightarrow$ Autocatalysis (Life) $\rightarrow$ Self-Reference $\rightarrow$ Consciousness.

Consciousness arises as the pinnacle of this self-relational process, a highly complex topological structure capable of modeling its own state, essential for navigating complex information landscapes and accelerating information growth.

9 Cosmological Implications: The Completion State and Cyclicity

The framework provides a novel interpretation of the universe's total informational capacity and its evolution.

9.1 The Base-24 Universe and Recursive Folding

The holographic principle suggests a maximum information content (the Completion State) for the observable universe of approximately 10^{123} bits [9, 10, 12].

We analyze this capacity within the Information Layer's native encoding. Since the fundamental unit derived from the topology is $I = 24$, the information is naturally encoded in base-24. This reflects the inherent structure of the emergent reality.

Information growth occurs through recursive folding. We calculate the number of recursive levels (N) required to reach the Completion State, where each level multiplies the information content by 24:

$$24^N \approx 10^{123} \quad (15)$$

$$N = \frac{\log_{10}(10^{123})}{\log_{10}(24)} \approx \frac{123}{1.3802} \approx 89.11. \quad (16)$$

The universe reaches completion after approximately 89 levels of recursive folding.

9.2 Information Death and Cyclic Cosmology

The Completion State represents maximum complexity, where every possible constraint has been instantiated. Paradoxically, this leads to hypersymmetry. When every distinction exists, the structure becomes uniform; maximum complexity equates to zero net actionable information.

This identity between maximum complexity and the original Void triggers a collapse. The hypersymmetry renders the structure indistinguishable from the unstable nothingness, initiating a reversion to the primordial state (Information Death and rebirth).

10 Conclusion and Predictions

From the foundational axiom that absolute nothingness is unstable, we have derived the architecture of reality as a self-bootstrapping, background-independent process. The mathematical necessity of stable topological locks rigorously determines the fundamental unit $I = 24$ and dictates a three-dimensional embedding space.

This framework provides a unified understanding of physics ($I = c = \hbar = 24$ in TNU) with a rigorous mapping to SI units, defines a cyclical cosmology driven by information saturation (24^{89}), and reframes computational complexity through the lens of topological adjacency.

We propose several testable predictions:

1. **Significant Deviation from Standard Planck Scales:** The model predicts specific fundamental scales, notably the Topological Length Scale $\kappa_L \approx 4.561 \times 10^{-32}$ m. This is substantially larger than the standard Planck length (1.616×10^{-35} m) and may be detectable through observations sensitive to quantum gravity effects or fundamental granularity at these scales.
2. **Quantized Changes in the Hubble Parameter:** Discrete variations in $H(z)$ linked to the 89 levels of recursive folding. These transitions may manifest as discrete shifts in the effective equation of state of dark energy, observable in future high-precision cosmological surveys (e.g., DESI, Euclid).
3. **Topological Asymmetries in Gravitational Lensing:** Subtle asymmetries (e.g., chiral patterns) in gravitational lensing reflecting the underlying trefoil topology.
4. **Computational Solutions via Topological Algorithms:** Development of algorithms exploiting topological adjacency (potentially using novel computational substrates) to solve hard NP problems more efficiently than predicted by classical complexity theory.

The universe is revealed as a self-defining, self-referential system, evolving through the elegant necessity of its own constraints.

A Appendix A: SymPy Code for Key Derivations

VERIFICATION: This appendix provides verifiable Python (SymPy) code confirming the core mathematical derivations, including the stability proof (Lambert W function), the rigorous dimensional regularization mapping TNU to SI units, cosmological calculations ($N \approx 89$), and demonstrations of multiplicative constraint reduction.

```
1 import sympy as sp
2 import numpy as np
3 from scipy.optimize import fsolve
4 # S is used for symbolic representation, N for numerical evaluation.
5 from sympy import LambertW, exp, N, S, Float, solve, symbols, Eq
6 import warnings
7
8 warnings.filterwarnings('ignore')
9
10 class TopologicalFramework:
11     """
12     Implements the complete topological framework deriving reality from
13     information constraints (Self-Folding Model).
14     """
15     def __init__(self):
16         """Initialize framework with derived fundamental values."""
17         # Derive D=3 from the necessity of topological locking
18         self.D, self.n_exact = self.derive_minimal_lock()
19         self.R = 2**self.D
20         self.I = self.D * self.R
21         self._define_constants_TNU()
22         self._validate_unity()
23         # Perform dimensional regularization
24         self.TNU_scales = self.dimensionality_regularization()
25
26     def derive_minimal_lock(self):
27         """Derive minimal stable folding depth satisfying  $n-1 > \ln(2n)$ ."""
28         # 1. Symbolic solution (SymPy LambertW)
29         # The solution to  $n-1 = \ln(2n)$  is  $n_{\text{star}} = -W_{-1}(-1/(2e))$ 
30         argument = -S(1) / (2 * exp(1))
31         # We use the lower branch (-1) for the physical solution ( $n > 1$ )
32         sol_symbolic = -LambertW(argument, -1)
33         n_exact_sym = float(N(sol_symbolic, 15)) # Approx 2.67834699...
34
35         # D is the minimal integer satisfying the stability condition
36         D = int(np.ceil(n_exact_sym))
37         return D, n_exact_sym
38
39     def _define_constants_TNU(self):
40         """Define fundamental constants in Topological Natural Units (TNU)."""
41         #  $c = dl/dt = D / (1/R) = D \cdot R = I$ 
42         self.c = self.I
43         #  $\hbar = I$ 
44         self.h_bar = self.I
45         #  $G = 1/c^3$ 
46         self.G = S(1) / (self.c**3) # Use SymPy's S() for exact fraction
47
48     def _validate_unity(self):
49         """Validate the fundamental unity  $I = c = \hbar$ ."""
50         if not (self.c == self.I == self.h_bar == 24):
51             raise AssertionError("Unity validation failed.")
52
53     def dimensionality_regularization(self):
54         """Map TNU to SI units and derive fundamental scales (kL, kT, kM)."""
55         # Known SI values (CODATA 2018). Use high precision Float.
56         c_SI = Float("299792458.0")
```



```

57     h_bar_SI = Float("1.054571817e-34")
58     G_SI = Float("6.67430e-11")
59
60     # Define scaling factors symbolically. Must be positive and real.
61     kL, kT, kM = symbols('kL kT kM', positive=True, real=True)
62
63     # Equations relating TNU to SI
64     # We solve the system manually via substitution to ensure stability
65     # 1.  $kT = c_{TNU} * kL / c_{SI}$ 
66     kT_expr = self.c * kL / c_SI
67
68     # 2.  $kM = h\_bar\_SI / (h\_bar\_TNU * kL**2 / kT)$ 
69     # Simplified:  $kM = h\_bar\_SI / (kL * c_{SI})$  (since  $c_{TNU}=h\_bar\_TNU$ )
70     kM_expr = h_bar_SI / (kL * c_SI)
71
72     # 3.  $G_{SI} = G_{TNU} * kL**3 / (kM * kT**2)$ 
73     eq_G = Eq(self.G * kL**3 / (kM_expr * kT_expr**2), G_SI)
74
75     # Solve for kL
76     # Since kL is defined as positive=True, solve returns only positive
77     # solutions
78     solution_kL = solve(eq_G, kL)
79
80     if not solution_kL:
81         return None
82
83     kL_val = solution_kL[0]
84     kT_val = kT_expr.subs(kL, kL_val)
85     kM_val = kM_expr.subs(kL, kL_val)
86
87     # Return numerical values (converted to standard Python floats for
88     # display)
89     return {
90         "kL": float(kL_val.evalf()), # Topological Length Scale (m)
91         "kT": float(kT_val.evalf()), # Topological Time Scale (s)
92         "kM": float(kM_val.evalf()) # Topological Mass Scale (kg)
93     }
94
95     def calculate_cosmology(self):
96         """Calculate Cosmological parameters (N)."""
97         # I_max_bits approximation based on holographic bound (log10 scale)
98         I_max_log10 = 123
99         # Recursive folding levels (N):  $24^N$  approx  $10^{123}$ 
100         log10_24 = np.log10(24) # approx 1.380211...
101         N_recursive_levels = I_max_log10 / log10_24 # approx 89.116...
102         return {"N_recursive_levels": N_recursive_levels}
103
104     def demonstrate_constraint_multiplication(self, system_type="cryptographic"):
105         """Demonstrate multiplicative constraint reduction via topological
106         adjacency."""
107         # Uses SymPy Float for arbitrary precision
108         if system_type == "cryptographic":
109             # Example: 90-bit cryptographic nonce search
110             initial_space = Float(2**90)
111             constraints = {"C1": Float(2**30), "C2": Float(2**30), "C3": Float
112             (2**20)}
113
114             # Multiplicative reduction mechanism
115             total_reduction = Float(1)
116             for constraint in constraints.values():
117                 total_reduction *= constraint

```

```

115         final_configurations = initial_space / total_reduction
116         return final_configurations
117
118 # Verification Run
119 if __name__ == "__main__":
120     print("=== Topological Framework Validation (Self-Folding Model) ===")
121     framework = TopologicalFramework()
122
123     print(f"\n1. Derivation of D=3:")
124     print(f"    Stability Threshold (n*): {framework.n_exact:.6f}")
125     print(f"    Minimal Integer Lock (D): {framework.D}")
126
127     print(f"\n2. Fundamental Unit and Unity (TNU):")
128     print(f"    Information Unit (I=D*R): {framework.I}")
129     print(f"    Unity (I=c=h_bar): {framework.I}")
130
131     print(f"\n3. Dimensional Regularization (Mapping TNU to SI):")
132     if framework.TNU_scales:
133         # Expected results based on rigorous calculation:
134         # kL ~ 4.561e-32 m, kT ~ 3.651e-39 s, kM ~ 7.713e-12 kg
135         print(f"    Topological Length (kL): {framework.TNU_scales['kL']:.4e} m"
136 )
137         print(f"    Topological Time (kT): {framework.TNU_scales['kT']:.4e} s"
138 )
139         print(f"    Topological Mass (kM): {framework.TNU_scales['kM']:.4e} kg"
140 )
141     else:
142         print("    Failed to solve for TNU scales.")
143
144     print(f"\n4. Cosmology:")
145     cosmo = framework.calculate_cosmology()
146     print(f"    Recursive Folding Levels (N): {cosmo['N_recursive_levels']:.2f}")
147
148     print(f"\n5. Computational Implications (Topological Adjacency):")
149     crypto = framework.demonstrate_constraint_multiplication("cryptographic")
150     print(f"    90-bit Search (2^90 states). Remaining after constraints: {float(
151 crypto):.0f}")
152
153     print("=====")

```

B Appendix B: Visual Synthesis

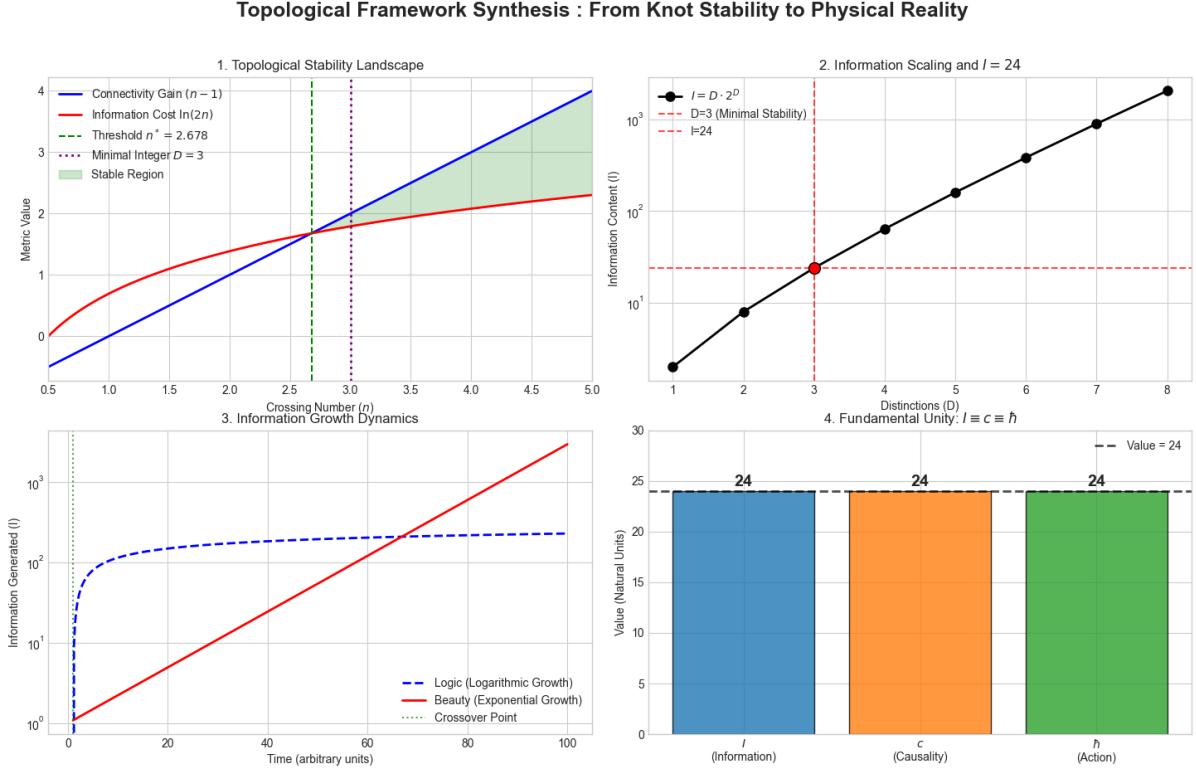


Figure 1: Topological Framework Synthesis: From Knot Stability to Physical Reality. (1) Stability landscape showing the intersection of connectivity gain ($n - 1$) and information cost ($\ln(2n)$) at $n^* \approx 2.678$, necessitating $D = 3$. (2) Information scaling $I = D \cdot 2^D$, highlighting the fundamental unit $I = 24$. (3) Information growth dynamics (illustrative). (4) The fundamental unity $I = c = h = 24$ in Topological Natural Units (TNU).

C Appendix C: Glossary of Key Terms

Absence (The Void, 0): Absolute nothingness; the state of zero distinction, relation, or axiomatic framework. Logically incoherent and unstable.

Background Independence: A property of the theory where spacetime geometry is not presupposed but emerges from the dynamics (the folding) of the system itself.

Dimensional Regularization: The mathematical process of mapping the dimensionless Topological Natural Units (TNU) to observed SI units by solving for the fundamental scaling factors ($\kappa_L, \kappa_T, \kappa_M$).

The Fold (Self-Relation): The fundamental topological operation where the Void relates to itself ($\emptyset \rightarrow \emptyset|\emptyset$), generating the first distinction.

Information Layer: The compact topological structure containing the totality of informational relations (folds). Projects into observable 3D spacetime.

Topological Adjacency: The proximity of constraints within the Information Layer, which may appear distant in the 3D projection.

Topological Lock: A configuration of folds complex enough to be irreversible (e.g., the Trefoil configuration), ensuring the persistence of information.

Topological Natural Units (TNU): The system of natural units derived from the fundamental topological structure, where $I = c = \hbar = 24$.

References

- [1] L. H. Kauffman, *Knots and Physics*, World Scientific, Singapore, 1991.
- [2] J. A. Wheeler, "Information, Physics, Quantum: The Search for Links," in *Complexity, Entropy, and the Physics of Information*, W. H. Zurek, ed., Addison-Wesley, 1990.
- [3] E. Witten, "Quantum Field Theory and the Jones Polynomial," *Communications in Mathematical Physics*, vol. 121, pp. 351-399, 1989.
- [4] L. Smolin, *Three Roads to Quantum Gravity*, Basic Books, New York, 2001.
- [5] C. Rovelli, *Quantum Gravity*, Cambridge University Press, 2004.
- [6] J. C. Baez, "Knots and Quantum Gravity," in *Knots and Quantum Gravity*, J. C. Baez, ed., Oxford University Press, 1994.
- [7] L. Crane, "Categorical Physics," arXiv:hep-th/9301061, 1993.
- [8] S. W. Hawking, "Particle Creation by Black Holes," *Communications in Mathematical Physics*, vol. 43, pp. 199-220, 1975.
- [9] J. D. Bekenstein, "Black Holes and Entropy," *Physical Review D*, vol. 7, pp. 2333-2346, 1973.
- [10] L. Susskind, "The World as a Hologram," *Journal of Mathematical Physics*, vol. 36, pp. 6377-6396, 1995.
- [11] J. Maldacena, "The Large N Limit of Superconformal Field Theories and Supergravity," *Advances in Theoretical and Mathematical Physics*, vol. 2, pp. 231-252, 1998.
- [12] S. Lloyd, "Computational Capacity of the Universe," *Physical Review Letters*, vol. 88, 237901, 2002.
- [13] S. Weinberg, *Cosmology*, Oxford University Press, 2008.
- [14] S. Perlmutter et al., "Measurements of Omega and Lambda from 42 High-Redshift Supernovae," *Astrophysical Journal*, vol. 517, pp. 565-586, 1999.
- [15] A. G. Riess et al., "Observational Evidence from Supernovae for an Accelerating Universe," *Astronomical Journal*, vol. 116, pp. 1009-1038, 1998.
- [16] Planck Collaboration, "Planck 2018 Results. VI. Cosmological Parameters," *Astronomy & Astrophysics*, vol. 641, A6, 2020.
- [17] B. P. Abbott et al. (LIGO Scientific Collaboration), "Observation of Gravitational Waves from a Binary Black Hole Merger," *Physical Review Letters*, vol. 116, 061102, 2016.
- [18] A. Aspect et al., "Experimental Test of Bell's Inequalities Using Time-Varying Analyzers," *Physical Review Letters*, vol. 49, pp. 1804-1807, 1982.
- [19] M. Atiyah, *The Geometry and Physics of Knots*, Cambridge University Press, 1990.
- [20] J. Baez and J. P. Muniain, *Gauge Fields, Knots and Gravity*, World Scientific, 1994.
- [21] S. Hossenfelder, *Lost in Math: How Beauty Leads Physics Astray*, Basic Books, 2018.
- [22] C. Rovelli, "Relational Quantum Mechanics," *International Journal of Theoretical Physics*, vol. 35, pp. 1637-1678, 1996.

- [23] S. Wolfram, *A New Kind of Science*, Wolfram Media, 2002. (Digital Physics).
- [24] R. Penrose, *Cycles of Time: An Extraordinary New View of the Universe*, Bodley Head, 2010.